

Exercise 01 : SQL Foundations

SELECT & Filtering

Selecting and sorting

1) Retrieve all columns

SELECT *

FROM employees;



ID	First-name	Last-name	Department	Salary	Hire date	City
1	Alice	Green	IT	70 000	2000-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	Makoreu	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72 000	2017-09-30	Johannesburg

2) Unique departments

SELECT DISTINCT department

FROM employees;



Department
IT
HR
Finance
Marketing

3) Names ordered by salary DESC

```
SELECT first_name, last_name  
FROM employees  
ORDER BY salary DESC;
```



First_name	Last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4) Top 3 highest-paid employees

```
SELECT id, first_name, last_name, salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3
```



ID	First_name	Last_name	salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

5) Employees in IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```



ID	first_name	Last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6) Finance with salary > 60 000

```
SELECT *
FROM employees
WHERE department = 'Finance'
AND salary > 60000;
```



ID	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65 000

7) HR or Marketing

```
SELECT *
FROM employees
WHERE department = "HR" OR department = "Marketing";
```



ID	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8) Not in IT

SELECT *

FROM employees

WHERE department $\neq$ ~~IT~~ 'IT';

ID	First_name	Last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9) IN IT, HR, Finance

SELECT *

FROM employees

WHERE department IN ('IT', 'HR', 'Finance');

ID	First_name	Last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10) IT + salary > 65 000 and Johannesburg

SELECT *

FROM employees

WHERE department = 'IT' AND salary > 65000 AND city =
Johannesburg;

ID	first_name	last_name	department	Salary	city
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg